Data Checking Report

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Figure 1

Full Dataset Table Summary

Acronyms and Definitions:

URM: Underrepresented in medicine

```
Two classifications (generalized):
  More inclusive (URM)
  Less inclusive (URM_AAMC)
```

OSCE: Objective Structured Clinical Examination

```
A type of performance-based test to evaluate clinical competence
A percentage from 0 to 100
Open to possible grading bias
```

USMLE: United States Medical Licensing Examination

```
Scored from 0 to 300 with a minimum of 194 to pass Step 1 and 209 to pass Step 2
(From pre-2020 grading standards)
```

Project Summary

The majority of the current literature surrounds average USMLE scores between URM students and those that are not underrepresented. The goal of this project to identify if there is any bias associated with students that are underrepresented in medicine (URM) and their OSCE scores.

Since we want to identify if there is bias, it would be more valuable to look at all URM students rather than URM_AAMC students to better isolate if there is any bias towards this group of students.

We are also going to summarize the average USMLE scores based on the URM classification and the average of the clerkship final grades based on the URM classification.

Data Irregularities

Definition and Classification of URM:

First, there are a total of 6 different columns that contain some classifier of race or ethnicity. The column titled `amcas_othethnicity` has no entries, so it will not be used in the analysis. The column titled `amcasEthnicityDesc` only includes information on students that identify as "Hispanic, Latino, or of Spanish origin", so we will not use this in the analysis. The columns titled `race_desc` and `race_cat_desc` do not contain a "Hispanic, Latino, or of Spanish origin" option, so they will not be used in the analysis. The column titled `additional_ethnicity` contains the majority of the options that would be used to classify ethnicity, so it will be included in the analysis. The column titled `additional_race` contains the majority of the options that would be used to classify race, so it will be included in the analysis.

Race and Ethnicity Variables

```
## $amcas_othethnicity
## logical(0)
##
## $amcasEthnicityDesc
```

```
## [1] ""
## [2] "Hispanic, Latino, or of Spanish origin"
## [3] "Not Spanish/Hispanic/Latino/Latina"
## [4] "Colombian"
## [5] "Cuban"
## [6] "Argentinean"
##
## $race_desc
##  [1] "Asian"                            "White"
##  [3] "African"                          ""
##  [5] "American Indian or Alaskan Native" "Black or African American"
##  [7] "Other"                            "African American"
##  [9] "Afro-Caribbean"                   "Japanese"
## [11] "American Indian or Alaska Native"
##
## $race_cat_desc
## [1] "Asian"                            "White"
## [3] "Black or African American"        ""
## [5] "American Indian or Alaska Native" "Other"
##
## $additional_ethnicity
##  [1] "Other Non-URM"
##  [2] "White"
##  [3] "Black or African American"
##  [4] "Asian Vietnamese"
##  [5] "Spanish/Hispanic/Latino/Latina"
##  [6] "Mexican, Mexican American, Chicano/Chicana"
##  [7] "American Indian or Alaska Native"
##  [8] "Asian Chinese"
##  [9] "Not Hispanic or Latino"
## [10] "Decline to Respond"
## [11] "Asian Japanese"
## [12] "Native Hawaiian"
## [13] ""
## [14] "Asian Filipino"
## [15] "Asian Other"
## [16] "Asian Indian"
## [17] "Asian Korean"
##
## $additional_race
##  [1] "Asian; White"
##  [2] "White"
##  [3] "Black or African American"
##  [4] "Asian"
##  [5] ""
##  [6] "Hispanic, Latino, or of Spanish origin; White"
##  [7] "Other; White"
##  [8] "Hispanic, Latino, or of Spanish origin"
##  [9] "Unknown"
## [10] "American Indian or Alaska Native; White"
## [11] "Asian; Black or African American"
## [12] "Other"
## [13] "American Indian or Alaska Native; Asian; White"
## [14] "Black or African American; White"
```

```
## [15] "Asian; Other; White"
## [16] "Amer Ind-Alaska Nat;Black or African Amer;White"
```

URM and URM_AAMC Counts and Percentages

There are two different URM categories in the data, URM and URM_AAMC. From the data, the only difference in the classification for the two categories is whether or not that student classifies as "Asian Vietnamese" for `additional_ethnicity`. To better capture the possible bias that we are interested in, we will use the URM data rather than the URM_AAMC data.

Difference between URM and URM_AAMC compared to ethnicity and race

```
##   URM URM_AAMC additional_ethnicity additional_race
## 1   1        0     Asian Vietnamese           Asian
## 2   1        0     Asian Vietnamese
## 3   1        0     Asian Vietnamese           Asian
## 4   1        0     Asian Vietnamese           Asian
## 5   1        0     Asian Vietnamese           Asian
## 6   1        0     Asian Vietnamese           Asian
## 7   1        0     Asian Vietnamese           Asian
## 8   1        0     Asian Vietnamese
```

OSCE Data:

When we are looking at the OSCE scores, we are only going to compare the CommScores and the PhysExamScores as these are the scores that may hold bias.

We will average the OSCE scores, but we will do this keeping the CommScores and the PhysExamScores separated.

Summary by OSCE scores

```
osce_data <- data %>%
  select(ResearchID, gender, URM, P1SprBates_CommScore, P1SprBates_PhysExamScore, P2FallRamos_CommScore

osce_data |>
    tbl_summary(type = list(
      P1SprBates_CommScore ~ "continuous",
      P2FallRamos_CommScore ~ "continuous",
      P2SpringSnelling_CommScore ~ "continuous"
    )) # When looking at the different OSCE cases, the Bates, Ramos, and Snelling cases are categorical
```

OSCE Case and Part	Score Average
P1SprBates_CommScore	91.59
P1SprBates_PhysExamScore	93.34
P2FallRamos_CommScore	94.81
P2FallRamos_PhysExamScore	94.94
P2FallPayne_CommScore	93.19
P2FallPayne_PhysExamScore	88.87
P2SpringHyde_CommScore	89.71
P2SpringHyde_PhysExamScore	85.34
P2SpringSnelling_CommScore	92.64

OSCE Case and Part	Score Average
P2SpringSnelling_PhysExamScore	85.30

OSCE Part	Average over All Cases
CommScore	92.39
PhysExamScore	89.56

The averages across the different OSCE cases are not the same, possibly due to lack of proper preparation towards the end of the semester (as shared in the initial meeting). However, the assumed difficulty of the different OSCE cases is presumed to be the same, so we will work with the overall average for the CommScore and the PhysExamScore, combining all OSCE cases together.

OSCE Variance

OSCE Case and Part	Statistic	Value
P1SprBates_CommScore	Variance	71.17
P1SprBates_CommScore	SD	8.44
P1SprBates_PhysExamScore	Variance	40.82
P1SprBates_PhysExamScore	SD	6.39
P2FallRamos_CommScore	Variance	56.65
P2FallRamos_CommScore	SD	7.53
P2FallRamos_PhysExamScore	Variance	38.93
P2FallRamos_PhysExamScore	SD	6.24
P2FallPayne_CommScore	Variance	65.27
P2FallPayne_CommScore	SD	8.08
P2FallPayne_PhysExamScore	Variance	38.66
P2FallPayne_PhysExamScore	SD	6.22
P2SpringHyde_CommScore	Variance	147.83
P2SpringHyde_CommScore	SD	12.16
P2SpringHyde_PhysExamScore	Variance	92.94
P2SpringHyde_PhysExamScore	SD	9.64
P2SpringSnelling_CommScore	Variance	71.65
P2SpringSnelling_CommScore	SD	8.46
P2SpringSnelling_PhysExamScore	Variance	143.54
P2SpringSnelling_PhysExamScore	SD	11.98

Using the averages for the scores across all OSCE cases, we can find the pooled variance and standard deviation for the CommScore portion of the OSCE and the PhysExamScore portion of the OSCE. The formula that was used to calculate the pooled variance is below, where $K = 5$ (since we are pooling 5 OSCE cases), n_i is the number of entries for the corresponding OSCE case, and s_i^2 is the observed variance for that OSCE case.

$$s_p^2 = \frac{\sum_{i=1}^{K}(n_i - 1)s_i^2}{\sum_{i=1}^{K}(n_i - 1)}$$

OSCE Part	Value
CommScoreVariance_Pooled	82.46
PhysExamScoreVariance_Pooled	70.89
CommScoreSD_Pooled	9.08

OSCE Part	Value
PhysExamScoreSD_Pooled	8.42

Clerkship Scores:

Instead of using the clinical grades for the clerkship scores, we will look at the final grades, as decided in the initial meeting.

The different courses for the clerkship scores are graded as follows:

```
0: Fail
1: Pass
2: High Pass
3: Honors
9: Pass with Remediation
```

Values and Labels for Clerkship Scores

```
## [1] NA  0  1  2  3  9
```

```
##  [1] "Label"                "Fail"                 "Pass"
##  [4] "High Pass"            "Honors"               "Pass with remediation"
##  [7] "Non-URM"              "URM"                  "Unknown"
## [10] "Non-URiM"             "URiM"                 "Non-MSTP student"
## [13] "MSTP student"         "Year 1"               "Year 2"
## [16] "Not Struggling"       "Borderline"           "Definite"
## [19] "Struggling"
```

First thing to note is the `AAC_Final_Course_Grade` and `RCC_Final_Course_Grade` columns as they both have 179 missing entries. For the remainder of the data exploration on the clerkship scores, we will ignore these as they will be covered in the missing section.

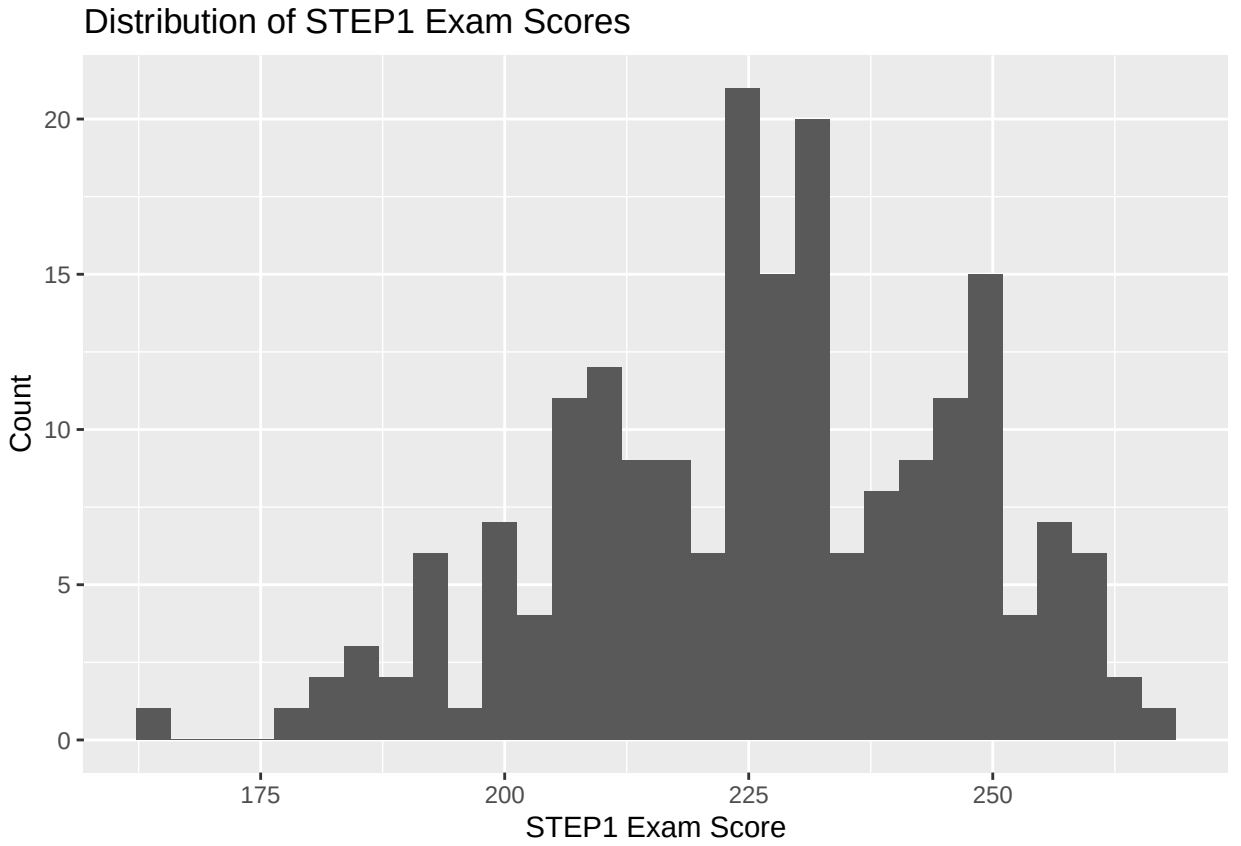
Both `EC_Final_Course_Grade` and `MSK_Final_Course_Grade` contain only passing grades, all entries are 1, with no missing entries for `EC_Final_Course_Grade` and 34 missing entries for `MSK_Final_Course_Grade`. Including these in the analysis will not provide any useful information, as all students scored the exact same in those courses, so we will not include them in the analysis of the clerkship scores.

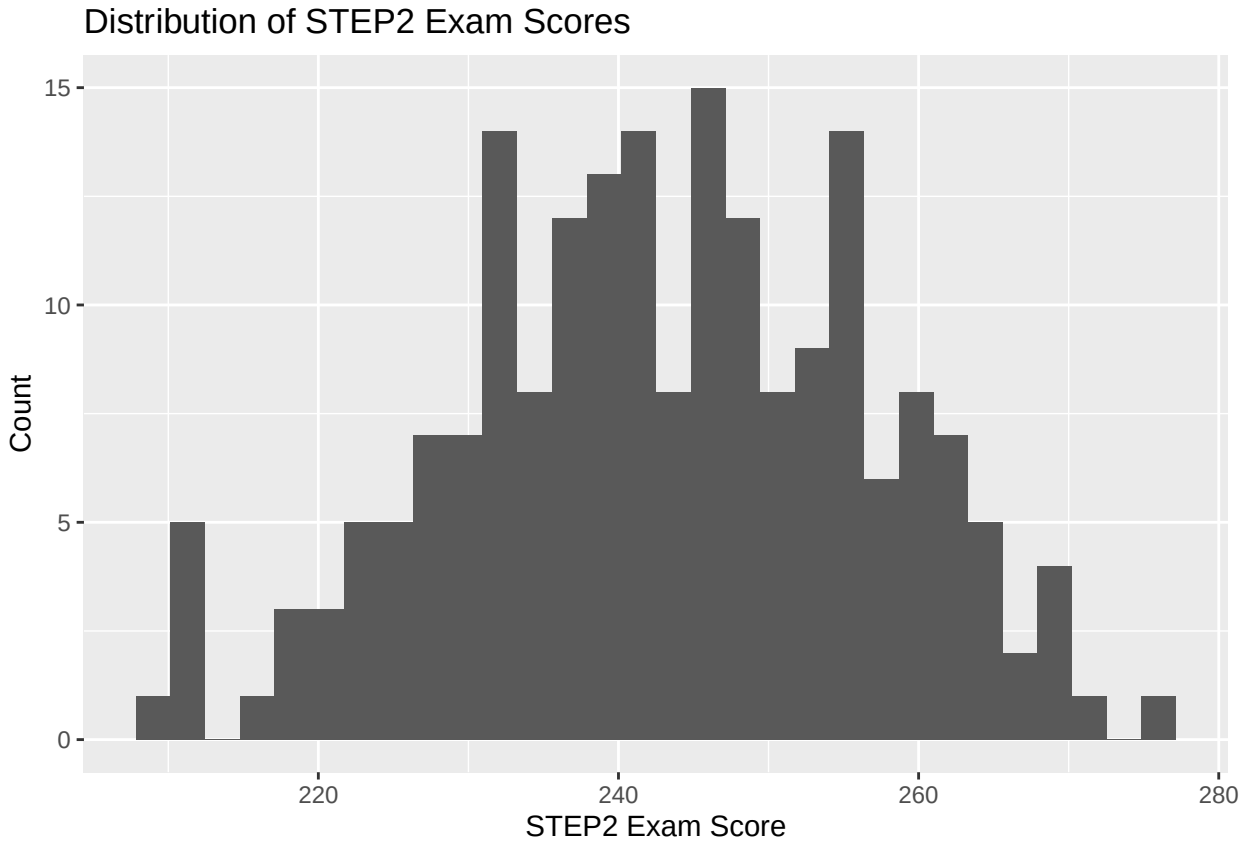
Looking at `OBG_Final_Course_Grade`, there is only one 9 recorded, and this is the only 9 for all of the final grades. Since 9 is pass with remediation, this student technically passed the course. So, we will include them in the pass group, 1, for the analysis.

USMLE Scores:

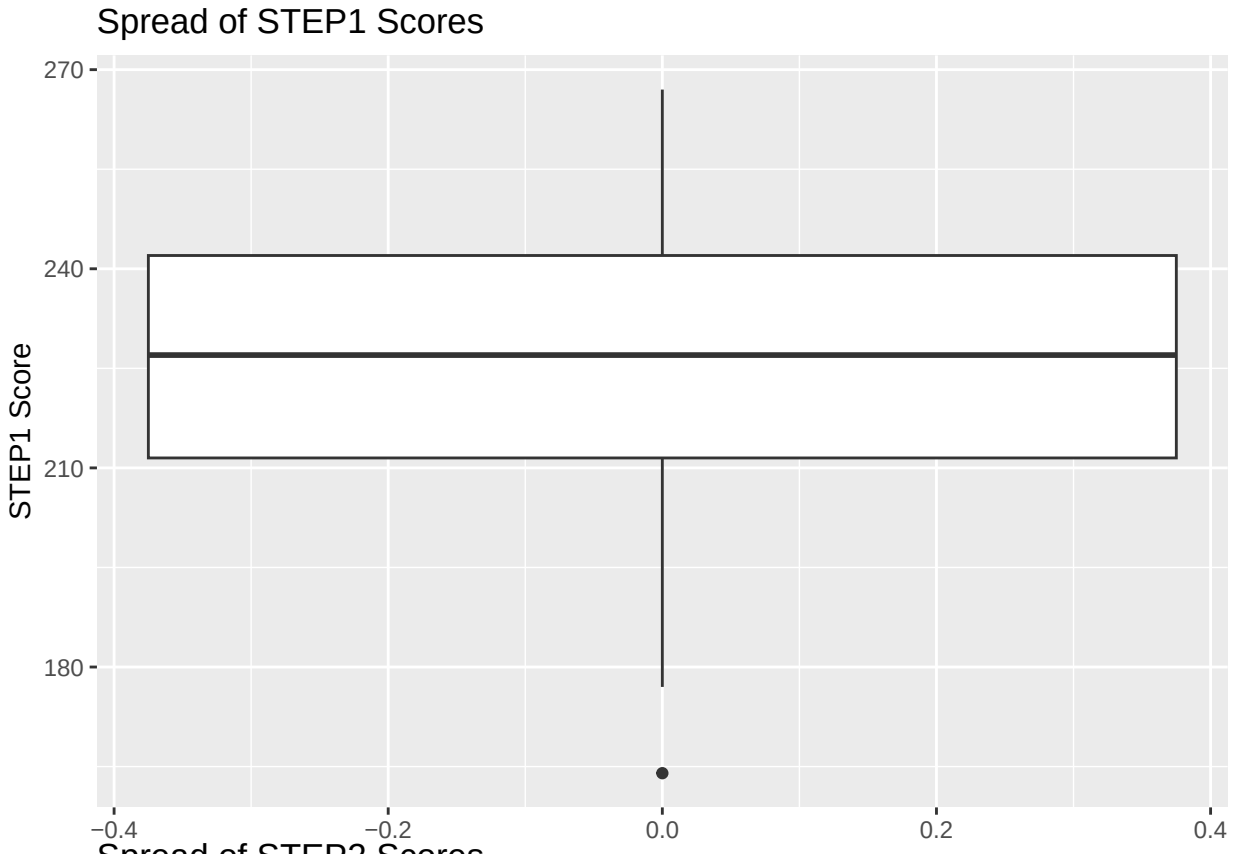
For the USMLE scores, we will only look at the final scores.

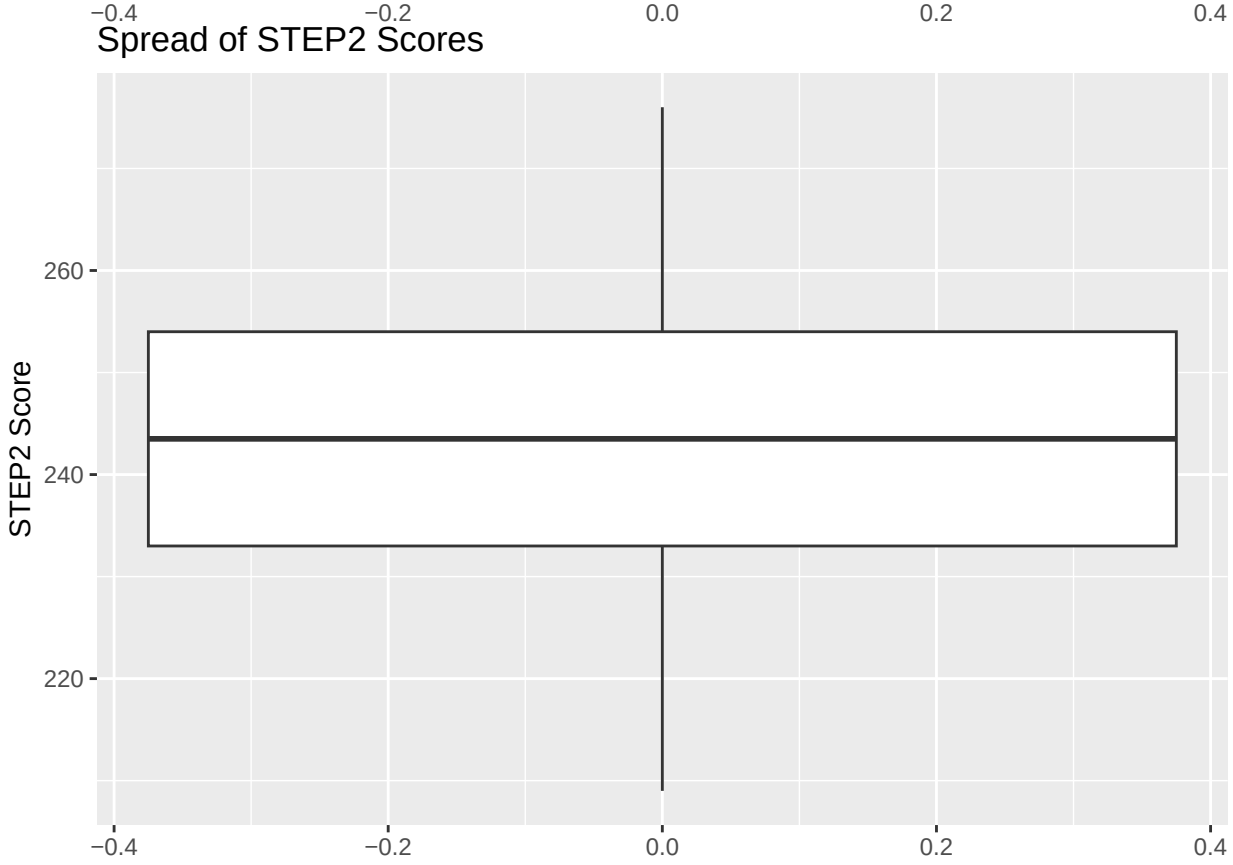
There is only 1 missing entry for the STEP2 Exam Score, and none for the STEP1 Exam Score.





Boxplots for STEP1 and STEP2 Exam Scores



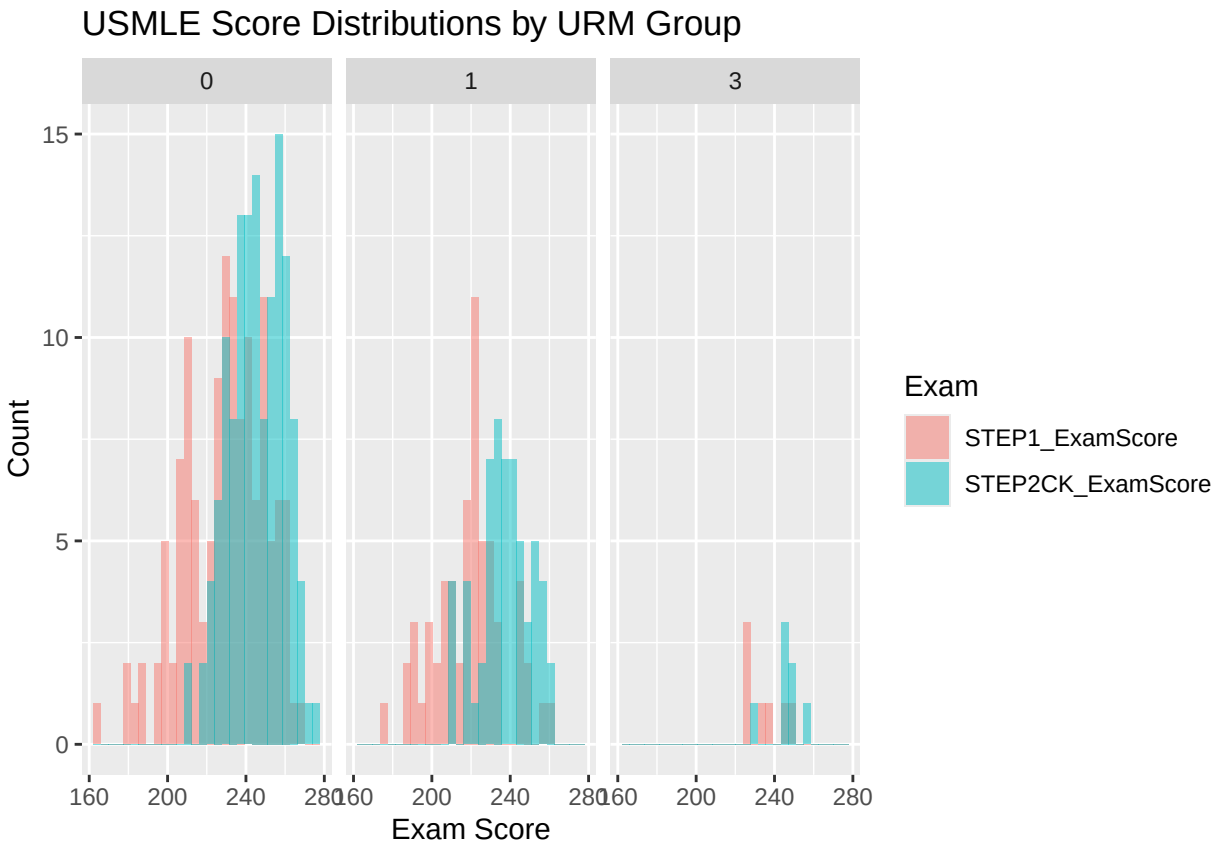


There are not any major issues that seem to arise looking at the USMLE scores. The scores for STEP2 are

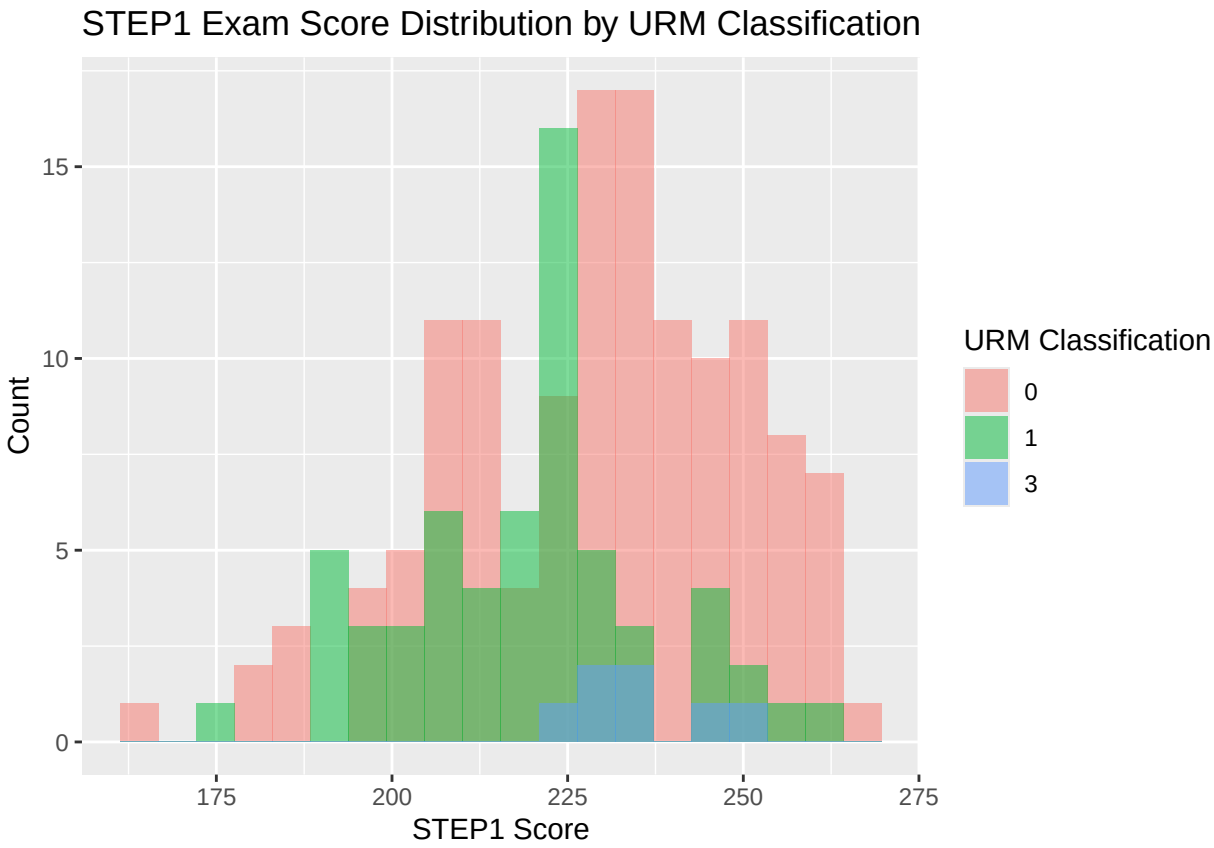
consistently higher than the score for STEP1, and this is seen regardless of URM classification.

Comparing Distributions of STEP1 and STEP 2 Exam Scores based on URM

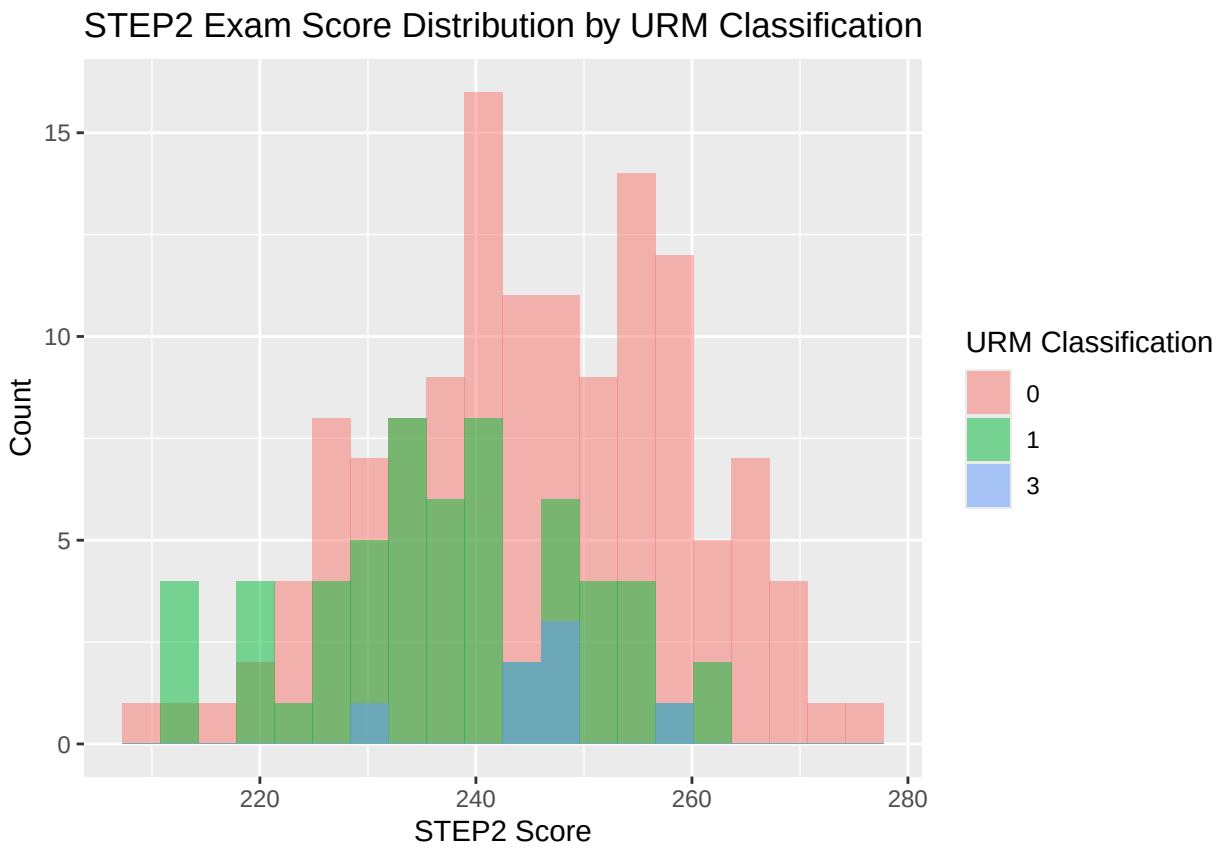
```
## Warning: Removed 1 row containing non-finite outside the scale range
## (‘stat_bin()‘).
```



Comparing the distribution of STEP1 scores based on the different URM classifications does not show anything significant in terms of possible issues in the data. The same can be said for comparing the distribution of STEP2 scores based on the different URM classifications.




```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_bin()`).
```



Missing Data

OSCE Data:

The missing entries for the CommScore and PhysScore is the same for each individual OSCE case.

Row by Row Comparison for Each OSCE Case

```
missing_osce <- osce_data %>%
  filter(
    is.na(P1SprBates_CommScore) != is.na(P1SprBates_PhysExamScore) |
    is.na(P2FallRamos_CommScore) != is.na(P2FallRamos_PhysExamScore) |
    is.na(P2FallPayne_CommScore) != is.na(P2FallPayne_PhysExamScore) |
    is.na(P2SpringHyde_CommScore) != is.na(P2SpringHyde_PhysExamScore) |
    is.na(P2SpringSnelling_CommScore) != is.na(P2SpringSnelling_PhysExamScore)
)
missing_osce
```

```
##  [1] ResearchID                    gender
##  [3] URM                           P1SprBates_CommScore
##  [5] P1SprBates_PhysExamScore      P2FallRamos_CommScore
##  [7] P2FallRamos_PhysExamScore     P2FallPayne_CommScore
##  [9] P2FallPayne_PhysExamScore     P2SpringHyde_CommScore
## [11] P2SpringHyde_PhysExamScore    P2SpringSnelling_CommScore
## [13] P2SpringSnelling_PhysExamScore
```

```
## <0 rows> (or 0-length row.names)
```

```
# From this point forward, any information on missing OSCE scores is going to be coded using only CommS
```

```
##              Total Missing     Missing From All Cases
##                         46                         39
##     Missing From Only Case Missing From More Than One
##                          2                          5
```

Over all of the OSCE cases, there are 153 students with no missing values, 39 with all missing values, and 7 with a combination of missing and not missing. So, a total of 46 students are missing some number of OSCE scores. There is around 23% of the students that are missing OSCE scores out of the total number of students.

$$\frac{\text{Number of Students Missing OSCE Scores}}{\text{Total Number of Students}} = \frac{46}{199} \approx 0.2312$$

Of these 46 that have missing scores, 17 are classified as URM, 28 as not URM, and 1 as unknown. Comparing percentages of the total missing grouped by URM classification to the total percentage of the URM classifications from section 3.1.1, we can see that the students that are missing OSCE scores do not seem to be from a specific URM classification. We can also see that the students that are missing OSCE scores are almost equally female and male, which is similar to the total sample.

```
##     Missing Classified as URM Missing Classified as Not-URM
##                            17                            28
##        Missing Not Classified
##                             1
```

```
##     Percentage of Total Missing Classified as URM
##                                             0.370
## Percentage of Total Missing Classified as Not-URM
##                                             0.609
##        Percentage of Total Missing Not Classified
##                                             0.022
```

```
##
##  F  M
## 25 21
```

```
## Percentage of Total Missing, Female   Percentage of Total Missing, Male
##                               0.543                               0.457
```

Comparing the averages of the non-missing OSCE scores the 7 students that have some combination of missing OSCE scores to the overall averages for those OSCE scores will show if removing these 7 students from the analysis will significantly change the data.

Research ID's and Corresponding Rows That Have 1-4 Missing OSCE Scores

```
## [1] 104091 105342 105765 106361 107294 107424 107995
```

```
## [1]  64 100 106 114 136 144 154
```

```
## [1] 104091 105342 105765 106361 107294 107424 107995
```

```
##    ResearchID P2SpringHyde_CommScore P2SpringSnelling_CommScore
## 64     104091                    95                         70
```

```
##     ResearchID P1SprBates_CommScore P2SpringHyde_CommScore
## 100     105342                  100                    100
##     P2SpringSnelling_CommScore
## 100                         95
```

```
##     ResearchID P2FallRamos_CommScore P2FallPayne_CommScore
## 106     105765                  100                92.86
##     P2SpringHyde_CommScore P2SpringSnelling_CommScore
## 106                     95                         80
```

```
##     ResearchID P1SprBates_CommScore P2FallRamos_CommScore P2FallPayne_CommScore
## 114     106361                   70                    95                    100
```

```
##     ResearchID P1SprBates_CommScore P2FallRamos_CommScore P2FallPayne_CommScore
## 136     107294                   80                    95                     85
```

```
##     ResearchID P2FallRamos_CommScore P2FallPayne_CommScore
## 144     107424                   95                96.43
##     P2SpringHyde_CommScore P2SpringSnelling_CommScore
## 144                    100                        100
```

```
##     ResearchID P1SprBates_CommScore P2FallRamos_CommScore P2FallPayne_CommScore
## 154     107995                   95                    85                     85
```

Table of Rows with More than One OSCE Score Missing and Table of Averages

OSCE Case and Part	Score Average
P1SprBates_CommScore	91.59
P1SprBates_PhysExamScore	93.34
P2FallRamos_CommScore	94.81
P2FallRamos_PhysExamScore	94.94
P2FallPayne_CommScore	93.19
P2FallPayne_PhysExamScore	88.87
P2SpringHyde_CommScore	89.71
P2SpringHyde_PhysExamScore	85.34
P2SpringSnelling_CommScore	92.64
P2SpringSnelling_PhysExamScore	85.30

```
##     ResearchID P1SprBates_CommScore P2FallRamos_CommScore P2FallPayne_CommScore
## 64      104091                   NA                    NA                     NA
## 100     105342                  100                    NA                     NA
## 106     105765                   NA                   100                  92.86
## 114     106361                   70                    95                 100.00
## 136     107294                   80                    95                  85.00
## 144     107424                   NA                    95                  96.43
## 154     107995                   95                    85                  85.00
##     P2SpringHyde_CommScore P2SpringSnelling_CommScore
## 64                      95                         70
```

```
## 100                    100                         95
## 106                     95                         80
## 114                     NA                         NA
## 136                     NA                         NA
## 144                    100                        100
## 154                     NA                         NA
```

```
##     ResearchID P1SprBates_PhysExamScore P2FallRamos_PhysExamScore
## 64      104091                      NA                        NA
## 100     105342                   91.18                        NA
## 106     105765                      NA                    100.00
## 114     106361                   90.00                     83.00
## 136     107294                   90.00                     92.00
## 144     107424                      NA                     96.15
## 154     107995                   96.00                     84.00
##     P2FallPayne_PhysExamScore P2SpringHyde_PhysExamScore
## 64                         NA                      73.86
## 100                        NA                      89.77
## 106                     85.51                      85.23
## 114                    100.00                         NA
## 136                     79.00                         NA
## 144                     94.20                      95.45
## 154                     88.00                         NA
##     P2SpringSnelling_PhysExamScore
## 64                           83.33
## 100                          98.55
## 106                          81.16
## 114                             NA
## 136                             NA
## 144                          98.55
## 154                             NA
```

Out of the seven students that have some number of missing OSCE scores but not all missing, the deviation of the students non-missing OSCE scores from the mean are between 0 and up to 2 standard deviations away from the mean. So, there are not any extremes in these seven students, meaning if they are excluded from analysis the overall data would not be altered in an extreme way.

```
##   ResearchID P1SprBates_CommScore P2FallRamos_CommScore P2FallPayne_CommScore
## 1     104091                   NA                    NA                    NA
## 2     105342             8.407643                    NA                    NA
## 3     105765                   NA             5.1898734            -0.3281013
## 4     106361           -21.592357             0.1898734             6.8118987
## 5     107294           -11.592357             0.1898734            -8.1881013
## 6     107424                   NA             0.1898734             3.2418987
## 7     107995             3.407643            -9.8101266            -8.1881013
##   P2SpringHyde_CommScore P2SpringSnelling_CommScore P1SprBates_PhysExamScore
## 1               5.286624                -22.643312                       NA
## 2              10.286624                  2.356688                -2.158471
## 3               5.286624                -12.643312                       NA
## 4                     NA                        NA                -3.338471
## 5                     NA                        NA                -3.338471
## 6              10.286624                  7.356688                       NA
## 7                     NA                        NA                 2.661529
```

```
##   P2FallRamos_PhysExamScore P2FallPayne_PhysExamScore
## 1                        NA                        NA
## 2                        NA                        NA
## 3                   5.05519                -3.3615823
## 4                 -11.94481                11.1284177
## 5                  -2.94481                -9.8715823
## 6                   1.20519                 5.3284177
## 7                 -10.94481                -0.8715823
##   P2SpringHyde_PhysExamScore P2SpringSnelling_PhysExamScore
## 1                -11.4757962                      -1.974777
## 2                  4.4342038                      13.245223
## 3                 -0.1057962                      -4.144777
## 4                         NA                             NA
## 5                         NA                             NA
## 6                 10.1142038                      13.245223
## 7                         NA                             NA
```

OSCE Case and Part	SD
P1SprBates_CommScore_SD	8.44
P1SprBates_PhysExamScore_SD	6.39
P2FallRamos_CommScore_SD	7.53
P2FallRamos_PhysExamScore_SD	6.24
P2FallPayne_CommScore_SD	8.08
P2FallPayne_PhysExamScore_SD	6.22
P2SpringHyde_CommScore_SD	12.16
P2SpringHyde_PhysExamScore_SD	9.64
P2SpringSnelling_CommScore_SD	8.46
P2SpringSnelling_PhysExamScore_SD	11.98

Clerkship Scores:

The percentage of missingness for the clerkship scores depends on the specific course.

The two courses that have a very high amount of missing entries are `AAC_Final_Course_Grade` and `RCC_Final_Course_Grade`, both with 179 missing entries. These two courses also have their entries in the exact same row, possibly showing that the name of the courses changed over the amount of time when the data was collected from the students.

The course `EC_Final_Course_Grade` has zero missing entries, so there is no need to check it alongside other courses.

The other courses have a missing rate of $7 - 46$, with the majority being around $\sim 20 - 30$. The course with 46 missing entries is `CPC_Final_Course_Grade`.

Table of Final Course Grade Counts and Percentages

```
## [1] 20
```

Comparing the missing rows of the other courses to the non-missing rows of `AAC_Final_Course_Grade`, we can see that `AAC_Final_Course_Grade` can fill in the missing gaps of `CPC_Final_Course_Grade` perfectly. However, that means that `RCC_Final_Course_Grade` can also fill in the missing gaps of `CPC_Final_Course_Grade` perfectly.

Comparing AAC to Other Courses

```
## [1] 20
```

```
## [1] 0
```

```
## [1] 0
```

```
## [1] 0
```

```
## [1] 0
```

```
## [1] 0
```

```
## [1] 0
```

```
## [1] 0
```

```
## [1] 20
```

Comparing `AAC_Final_Course_Grade` and `RCC_Final_Course_Grade`, we can see that the two columns are not the same. So, unless there is new information on if a course name changed over time, we cannot merge any of these columns together.

Comparing Entries of AAC to Entries of RCC

```
##  [1] 2 1 2 1 3 2 2 2 3 3 1 2 2 2 2 2 1 1 2 3
```

```
##  [1] 3 2 2 1 3 1 2 2 2 2 1 1 3 2 2 1 1 2 2 3
```

Assuming there is no further information regarding the possibility of merging `CPC_Final_Course_Grade` and `AAC_Final_Course_Grade` or `CPC_Final_Course_Grade` and `RCC_Final_Course_Grade`, for the analysis we will ignore `AAC_Final_Course_Grade` and `RCC_Final_Course_Grade` and not include them in the analysis of the clerkship grades.

There are only a total of 40 students that have all of the final grades for the remaining courses, with the other 159 students having a combination of $1 - 7$ final grades missing from the 9 total courses.

```
## [1] 2 1 0 4 3 7
```

```
## [1] 40
```

```
## [1] 159
```

The percentage of students that have missing final grades that are classified as URM out of the total number of students missing final grades is ~28.9%. Comparing this to the total percent of URM students, 30%, we can assume that URM classification does not influence which students are missing final grades.

```
##     Missing Classified as URM Missing Classified as Not-URM
##                            46                            107
##        Missing Not Classified
##                             6
```

```
##     Percentage of Total Missing Classified as URM
##                                             0.289
## Percentage of Total Missing Classified as Not-URM
##                                             0.673
##        Percentage of Total Missing Not Classified
##                                             0.038
```

Since the majority of the missing entries are resulting from the `CPC_Final_Course_Grade` (the column that might be able to combine with `AAC_Final_Course_Grade` or `RCC_Final_Course_Grade`), we will ignore that column during the analysis as we do not have the full information on this course and the percent that is missing is almost 25% of all the data.

Overlapping Missing Clerk Scores

```
##   total CPC_miss HAC_miss ICAC_miss NC_miss OBG_miss OPC_miss PC_miss CPC_HAC
## 1   199       46       20        27       7       28       20      26       1
##   CPC_ICAC CPC_NC CPC_OBG CPC_OPC CPC_PC HAC_ICAC HAC_NC HAC_OBG HAC_OPC HAC_PC
## 1        2      0       3       2      1        2      2       3       2      3
##   ICAC_NC ICAC_OBG ICAC_OPC ICAC_PC NC_OBG NC_OPC NC_PC OBG_OPC OBG_PC OPC_PC
## 1       2        4        2       3      3      1     4       4      3      2
```

USMLE Scores:

Thankfully, the USMLE scores have almost no missing entries. However, there is one missing entry for the STEP2 scores. Since there is a large sample size to begin with, $n = 199$, comparing the average for STEP1 with all the entries and STEP2 with all but one of the entries, $n = 198$, will not result in a significant difference in the analysis. So, we will drop this one missing entry from STEP2 for the analysis.

Conclusions

Changes to Overall Data:

1. For race and ethnicity, we are only using the additional_race and additional_ethnicity columns

Why?: The other columns that contain data for race and ethnicity contain little to no useful information

2. We are using URM instead of URM_AAMC

Why?: URM flags all students as underrepresented in medicine that URM_AAMC does, except URM flags for "Asian Vietnamese" and URM_AAMC does not. URM will be a better indicator for a possible bias

Changes to OSCE Data:

1. Only looking at CommScore and PhysExamScore

Why?: The CommScore and PhysExamScore are the two portions that are open to grading bias

2. Averaging CommScore and PhysExamScore across the different OSCE cases

Why?: The assumed difficulty of the different cases is predicted to be the same, so averaging will give a better estimate

3. Dropping 46 total missing entries

Other Missing Information Based on OSCE Missing

```
##
##    1    2    3 <NA>
##    8   10    5   23
```

```
##
##    1    2    3 <NA>
##   15   18   10    3
```

```
##
##    1    2    3 <NA>
##   16   16   11    3
```

```
##
##  1  2  3
## 18 15 13
```

```
##
##    1    2    3 <NA>
##   17   14   11    4
```

```
##
##    1    2    3 <NA>
##   15   19   10    2
```

```
##
##    1    2    3 <NA>
##   15   15   13    3
```

Why?: If we drop all 46 students that have missing OSCE scores from the entire data set, we will be losing data for all of the other variables we are planning on analyzing. So, we should instead drop the 46 missing entries so that data is not unnecessarily thrown out. By dropping all 46, we are keeping the sample size the same for each OSCE, which will allow us to pool to means and variances of the different cases.

Changes to Clerkship Scores:

1. Ignoring EC and MSK Final Grade columns

Why?: All students passed these courses with no difference in performance

2. Combining “Pass” and “Pass with Remediation”

Why?: Pass with remediation only occurs once in the entire dataset, and this student still passed the course

3. Ignoring AAC, RCC, and CPC columns (Unless there is new information on the course names possibly changing)

Why?: There are only 20 entries for each AAC and RCC with the remaining entries as missing. Comparing to CPC suggests that there was a course name change as the 20 entries for AAC and RCC slot perfectly into the missing entries of CPC.If there is a verified course name change, we can merge the columns together. Otherwise, removing these columns from the analysis will significantly increase the sample size after accounting for missing data, as these three columns have the highest rate of missingness. This could possibly result in a biased analysis, however this is unlikely considering the courses have similar spreads of scores.

4. Dropping missing entries for each column

Why?: From the initial meeting, the reason for missing entries for the courses and the final grades is due to poor tracking. The percentage of missing for each course is consistent with this assumption as the majority of the missing percentages for the courses are $\sim 20\% - 30\%$.

Changes to USMLE Scores:

1. Averaging STEP1 with $n = 199$ and STEP2 with $n = 198$

Why?: Averaging with one less data point from a sample size that is already large will not alter the analysis

2. Removing missing entry for further analysis

Why?: There is only one missing entry, so this will decrease the sample size from $n = 199$ to $n = 198$ which result in insignificant change (if any) to the analysis.

Final Cleaned Data Set

OSCE Data:

Removing Missing Entries for OSCE Scores

```
## [1] 153
```

After removing the missing entries on the OSCE data, we are left with $n = 153$ for further analysis on OSCE scores.

Clerkship Data:

Removing Missing Entries for Clerkship Scores and Combining "Pass" and "Pass with Remediation"

```
## [1] 93
```

After removing the missing entries for all of the course final grades, we are left with $n = 93$ for further analysis on the clerkship scores. This is less than half the size of the original sample size. This is because there students missing grades for some courses but not others, with the reason for the missing data being poor recording, as discussed in the initial meeting.

USMLE Data:

Removing Missing Entry for USMLE Scores

```
## [1] 198
```

After removing the missing entry for the USMLE scores, we are left with $n = 198$ for further analysis on USMLE scores.

Appendix

Characteristic	N = 199[1]
ResearchID	101,098,109,962, (105,635)
Unknown	0 (0%)
AAC_clinical_grade	
1	5, (26%)
2	8, (42%)
3	6, (32%)
Unknown	180 (90%)
CPC_clinical_grade	
1	17, (13%)
2	45, (34%)
3	71, (53%)
Unknown	66 (33%)
HAC_clinical_grade	
1	41, (23%)
2	62, (35%)
3	73, (41%)
Unknown	23 (12%)
ICAC_clinical_grade	
1	40, (23%)
2	67, (39%)
3	64, (37%)
Unknown	28 (14%)
NC_clinical_grade	
1	16, (9.9%)
2	87, (54%)
3	58, (36%)
Unknown	38 (19%)
OBG_clinical_grade	
1	40, (27%)
2	51, (34%)
3	59, (39%)
Unknown	49 (25%)
OPC_clinical_grade	
1	21, (13%)
2	70, (44%)
3	67, (42%)
Unknown	41 (21%)
PC_clinical_grade	
1	49, (32%)
2	51, (34%)
3	52, (34%)
Unknown	47 (24%)
RCC_clinical_grade	
1	5, (26%)
2	7, (37%)
3	7, (37%)
Unknown	180 (90%)
AAC_Final_Course_Grade	
1	5, (25%)
2	11, (55%)

Characteristic	N = 199[1]
URM	
0	132, (66%)
1	60, (30%)
3	7, (3.5%)
URM_AAMC	
0	140, (70%)
1	52, (26%)
3	7, (3.5%)

[1] n, (%)

Characteristic	N = 199[1]
ResearchID	105,342 (103,730, 107,568)
gender	
D	1 (0.5%)
F	101 (51%)
M	97 (49%)
URM	
0	132 (66%)
1	60 (30%)
3	7 (3.5%)
P1SprBates_CommScore	95 (90, 100)
Unknown	42
P1SprBates_PhysExamScore	95.6 (91.2, 97.1)
Unknown	42
P2FallRamos_CommScore	100 (90, 100)
Unknown	41
P2FallRamos_PhysExamScore	96.2 (92.3, 100.0)
Unknown	41
P2FallPayne_CommScore	96 (89, 100)
Unknown	41
P2FallPayne_PhysExamScore	89.9 (85.5, 94.2)
Unknown	41
P2SpringHyde_CommScore	95 (80, 100)
Unknown	42
P2SpringHyde_PhysExamScore	88 (81, 92)
Unknown	42
P2SpringSnelling_CommScore	95 (90, 100)
Unknown	42
P2SpringSnelling_PhysExamScore	88 (81, 94)
Unknown	42

[1] Median (Q1, Q3); n (%)

Characteristic	**N = 199**[1]
ResearchID	105,342 (103,730, 107,568)
gender	
D	1 (0.5%)
F	101 (51%)
M	97 (49%)
URM	
0	132 (66%)
1	60 (30%)
3	7 (3.5%)
AAC_Final_Course_Grade	
1	5 (25%)
2	11 (55%)
3	4 (20%)
Unknown	179
CPC_Final_Course_Grade	
1	52 (34%)
2	50 (33%)
3	51 (33%)
Unknown	46
EC_Final_Course_Grade	
1	199 (100%)
HAC_Final_Course_Grade	
1	46 (26%)
2	71 (40%)
3	62 (35%)
Unknown	20
ICAC_Final_Course_Grade	
1	44 (26%)
2	66 (38%)
3	62 (36%)
Unknown	27
MSK_Final_Course_Grade	164 (99%)
Unknown	34
NC_Final_Course_Grade	
1	71 (37%)
2	56 (29%)
3	65 (34%)
Unknown	7
OBG_Final_Course_Grade	
1	50 (29%)
2	60 (35%)
3	60 (35%)
9	1 (0.6%)
Unknown	28
OPC_Final_Course_Grade	
1	41 (23%)
2	66 (37%)
3	72 (40%)
Unknown	20
PC_Final_Course_Grade	

Characteristic	**N = 199**[1]
ResearchID	105,635 (2,542)
gender	
D	1 (0.5%)
F	101 (51%)
M	97 (49%)
URM	
0	132 (66%)
1	60 (30%)
3	7 (3.5%)
STEP1_ExamScore	226 (20)
STEP2CK_ExamScore	243 (14)
Unknown	1

[1]Mean (SD); n (%)

Characteristic	**N = 199**[1]
gender	
D	1, (0.5%)
F	101, (51%)
M	97, (49%)

[1]n, (%)

Characteristic	N = 199[1]
AAC_Final_Course_Grade	
1	5 (25%)
2	11 (55%)
3	4 (20%)
Unknown	179
CPC_Final_Course_Grade	
1	52 (34%)
2	50 (33%)
3	51 (33%)
Unknown	46
EC_Final_Course_Grade	
1	199 (100%)
HAC_Final_Course_Grade	
1	46 (26%)
2	71 (40%)
3	62 (35%)
Unknown	20
ICAC_Final_Course_Grade	
1	44 (26%)
2	66 (38%)
3	62 (36%)
Unknown	27
MSK_Final_Course_Grade	164 (99%)
Unknown	34
NC_Final_Course_Grade	
1	71 (37%)
2	56 (29%)
3	65 (34%)
Unknown	7
OBG_Final_Course_Grade	
1	50 (29%)
2	60 (35%)
3	60 (35%)
9	1 (0.6%)
Unknown	28
OPC_Final_Course_Grade	
1	41 (23%)
2	66 (37%)
3	72 (40%)
Unknown	20
PC_Final_Course_Grade	
1	55 (32%)
2	62 (36%)
3	56 (32%)
Unknown	26
RCC_Final_Course_Grade	
1	6 (30%)
2	10 (50%)
3	4 (20%)
Unknown	179